IOP Conference Series: Earth and Environmental Science

PAPER • OPEN ACCESS

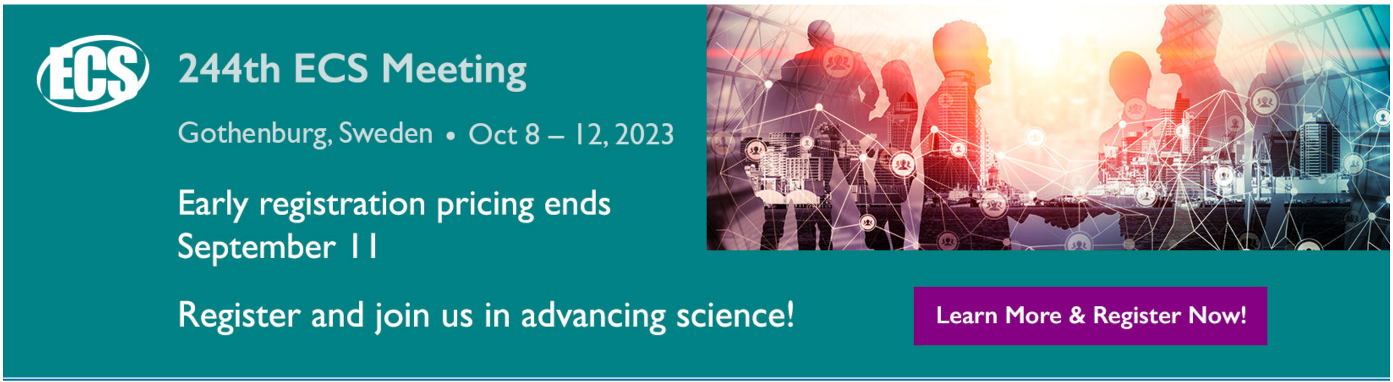
Feeding frequency on survival, growth, blood glucose of glass eel, *Anguilla bicolor bicolor* cultured in a recirculating aquaculture system

To cite this article: E Setiadi *et al* 2023 *IOP Conf. Ser.: Earth Environ. Sci.* **1221** 012037

View the article online for updates and enhancements.

You may also like

- The Use of Different Diets for Feeding Rate and Growth of Shortfin Eel (*Anguilla bicolor bicolor*)
  N Taufiq-Spj, S Sunaryo, A Wirasatria et al.
- Water and land efficiency in eel (*Anguilla bicolor bicolor*) rearing in development of urban aquaculture through vertical aquaculture system
  Eko Harianto, Eddy Supriyono, Tatag Budiardi et al.
- Histological investigation on gonad maturation of cultured short fin eel, *Anguilla bicolor* (mcclelland, 1844) in captivity.
  F F Ching, A Masli and S Senoo



244th ECS Meeting

Gothenburg, Sweden • Oct 8 – 12, 2023

Early registration pricing ends September 11

Register and join us in advancing science!

Learn More & Register Now!

This content was downloaded from IP address 78.138.47.190 on 18/08/2023 at 01:26

Feeding frequency on survival, growth, blood glucose of glass eel, *Anguilla bicolor bicolor* cultured in a recirculating aquaculture system

E Setiadi[1,*], I Taufik, L Setijaningsih, I Ardi, and D Puspaningsih

[*]Research Center for Fisheries, National Research and Innovation Agency, Republic of Indonesia, 16911 Bogor, West Java, Indonesia

*erisetiadi@yahoo.com

Abstract. Indonesian shortfin eel, *Anguilla bicolor bicolor* is one of the eel species that high market demand both for local and export. Feeding frequency is an important that related to production for culture. The purpose of this experiment is to determine the optimal feeding frequencies on survival and growth of shortfin eel at glass eel to elver stages cultured in a recirculating aquaculture system. Four treatments of different feeding frequencies and three replicates each were performed. The treatments were as followed A) 1 time a day; B) 2 times a day; C) 3 times a day; and D) 4 times a day. The result showed that the feeding frequency affect the survival, length and weight growths, and glucose value. The best of survival rate (82.61±3.15%), weight (1.91±0.28 g) and length (10.64±0.45cm) found at feeding frequency 3 times a day compared to the others (P<.0.05). Blood glucose value increases with increasing feeding frequency. Water quality such as dissolved oxygen (DO), temperature and pH were within the optimal range. Three times a day of feeding frequency is recommended for applying on glass eel culture in a recirculating aquaculture system.

Keywords: Feeding frequency, survival, growth, blood glucose, *Anguilla bicolor bicolor*

1. Introduction

Shortfin eel, *Anguilla bicolor bicolor* is a catadromous fish where fish movements from aquatic habitats that different salinity (freshwater, brackish water, and seawater) [1]. Indonesia has six eels belong to *Anguilla* species are *A. bicolor bicolor, A. marmorata, A. borneensis, A. celebesensis, A. interiosis* and *A. nebulosa* [2] where the famous one is *A. bicolor bicolor* due to high market both local and export commodities [3]. *A. bicolor bicolor* is a carnivorous fish that containing high nutrients and delicious meat. Thus, this species has been widely cultured by eel farmers in Indonesia.

Growth rate of *A. bicolor bicolor* from the glass eel (0.09 g in body weight) to consumption size (250 g in body weight) needs at least nine months for culture period [4]. The main obstacle of rearing this species is high mortality occurs on the glass eel stage and the survival rate under 20%. Thus, this condition affects the next culture such as elver to fingerling stages and fingerling stage to consumption size cannot be done [5]. Therefore, the research is focused on increasing growth and survival of fish is needed.

Content from this work may be used under the terms of the Creative Commons Attribution 3.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

Some experimets in term of feeding frequencies affects fish growth and survival has been reported such as Asian seabass, *Lates calcarifer* [6], largemouth bass, *Micropterus salmoides* [7], Tilapia, *Oreochromis niloticus* [8; 9], common carp, *Cyprinus carpio* [10], and on ornamental fish namely angelfish, *Pterophyllum scalare* and severum, *Heros severus* [11]. The blood glucose can be used as indicator of stress condition [12], however, feeding frequency has also influenced of blood glucose [8]. The research of feeding frequency on glass eel, *A. bicolor bicolor* culture has not been reported yet. Therefore, the purpose of this experiment is to determine the optimal feeding frequency on growth, survival, and profile of blood glucose of *A. bicolor bicolor* from glass eel to elver stages for three months culture period in a recirculating aquaculture system.

2. Materials and methods

2.1. Time and site

The research was conducted from June – October 2021 in Multi Species Hatchery at Research Station for Environment, Technology, and Toxicology of Freshwater aquaculture, Cibalagung Bogor, West Java, Indonesia

2.2. Container test.

The recirculation unit consisted of 12 of circular fiber tank with water volume of 200 L each were used. Recirculation unit was completed by aeration, water heater, sediment tank, filter tanks (physical, chemical, and biological filters), and UV light. Ground water was used. The system was operated for 14 days before used to remove the toxic substances.

2.3. Fish test and rearing

Fish test is glass eels, *A. bicolor bicolor* with 0.13±0.03g in body weight. The glass eels were obtained from farmer at Pelabuhan Ratu, Suka Bumi, west Java, Indonesia. The glass eels were adapted for 14 days before used the experiment. After that, the 1250 of glass eels were introduced into circular fiber tank with water volume of 200L. the Glass eels were cultivated for three months. Feeding rate was 5%/body weight/biomass. Glass eels were fed artificial diet. Feeding frequency were applied depends on the treatments. The different of feeding frequency as a treatment was followed A) one time per days; B) two times per day; C) three times per day; and D) four times per day. Each treatment consisted of three replications. The time of feeding of A treatment (09:00am), B treatment (09:00am and 03:00pm), C treatment (09:00am, 12:00am, and 03:00pm), and D treatment (09:00 am, 11:00 am, 01:00pm and 03:00pm). At 08:00 am, the glass eels were fed by nauplii Artemia with the density of 2 individuals/ml for 10 days culture period. Glass eels were cultured based on the guidelines of the Institutional Animal Care and Use Committee (IACUC, 2018) and have ethical approval 029/KEH/SKE/IX/2022 from the animal ethic committee of the School of Veterinary and Biomedical, IPB University.

2.4. Data parameters

15 of glass eels were sampled every 10 days to measurement the length and weight. The parameters of survival rate, absolute length and absolute weight were calculated using formula [13] were as followed:

Survival rate (SR)

$$SR = (Nt/No) \times 100 \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (1)$$

where
SR = survival rate (%)
Nt = Number of fish at the end of experiment
No= Number of fish at the initial of experiment

Absolute length

$$L = Lt- Lo \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (2)$$

Where
L = Absolute length (cm)
Lt= Final length (cm)
Lo= Initial length (cm)

Absolute weight

$$W = Wt-Wo \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (3)$$

Where
W= Absolute weight (g)
Wt= Finasl weight (g)
Wo= Initial weight (g)

The blood glucose measurement was conducted at the end of the experiment. Three of eels were taken for each tank and then transferred into backer glass with water volume of 0.5L, the water in the backer glass was dropped (three drops) using anesthetized “Ocean Free Stabilizer”. After that, blood glucose measured by using Sinocare Blood Glucose Meter. The blood (0.5 ml) was taken using syringe and then dropped to the kit then through the blood glucose meter. Water quality parameters such as water temperature, Dissolved oxygen (DO), and pH were measured using water checkers where ammonia (NH_3-N), nitrite (NO_2-N), nitrate (NO_3-N), phosphate (PO4-P), and alkalinity were analyzed at laboratory using spectrophotometer [14].

2.5. Data analysis

All data were analyzed using one way ANOVA (SPSS version 26 statistic program). If there is significant different among the treatments, post hoc multiple comparison of Duncan test was performed.

3. Results and discussion

3.1. Survival rate and growth (absolutes length and absolute weight)

The result of glass eel cultured at different feeding frequency shows that at feeding frequency three times per days is the highest (82.61±3.15%), followed by four times (77.47±3.37%), two times (57.31±4.80%), and one time (17.04±3.47%) per day, respectively (Figure 1).

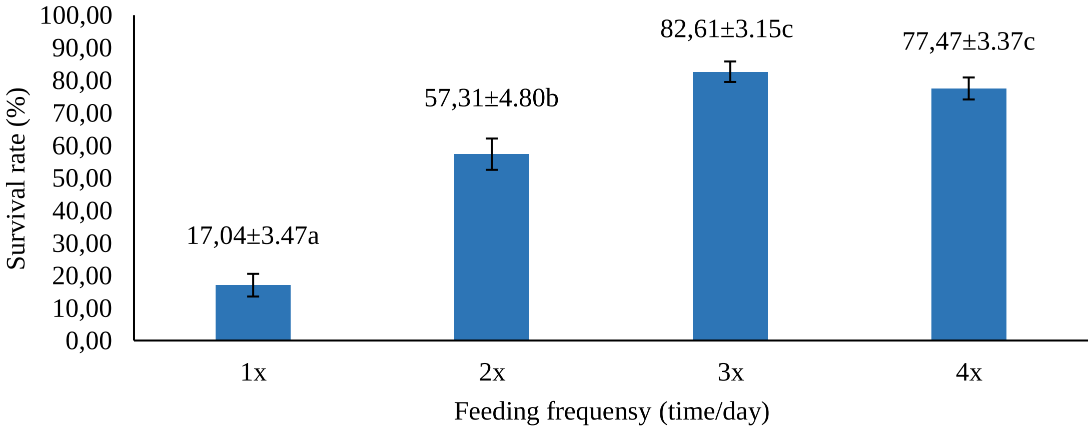


Figure 1. The survival rate of glass eel cultured at different feeding frequencies. Bars followed by the same superscript letters are not significantly different (P>0.05).

Statistical analysis revealed that three times a day feeding frequency is significantly different ($P<0.05$) in terms of survival rate compares to the others treatment, except for four times a day. This indicated that the feeding frequency affect the survival rate. The same result has been reported on Asian seabass, L*ates calcarifer* that three times feeding frequency is better than that of four times a day [6] while two times a day of feeding frequency is the best for largemouth bass, *Micropterus salmoides* [7]. The different of feeding frequency on survival at different fish species may be due to different of feeding habit and physiological. The present experiment shows that feeding frequency of one a day is the lowest. This condition may be due to long fasting to get next food which influence on physiological. All animals must eat to obtain energy for survive and growth, and other activities such as locomotion and reproduction. Physiological status of fish via hematological is a reliable indicator, especially, total plasma proteins. Total plasma proteins are the main components in term of diet metabolism and is particularly important in detecting alterations in protein metabolism [15]. The total blood protein levels are significantly decrease in lebranche mullet, *Mugil liza* at once a day of feeding frequency [16]. At the present experiment, one time feeding frequency has results in malnutrition due to lack of food intake, thus, such condition affects the fish survive. Based on the observation during the fish culture, the mortality of glass eel is frequently occurring at one time a day feeding.

The absolute length of glass eel cultured at different feeding frequency shows that feeding frequency of three times a day is the highest (10.64±0.45cm), followed by four times (9.45±0.13cm), three times (8.95±0.27cm), and one time (8.61±0.14cm), respectively (Figure 2). Statistical analysis revealed that three times a day of feeding frequency shows significantly different ($P<0.05$) compares to the others on absolute length. This indicated that feeding frequency influence on absolute length growth. Food is necessary for fish growth and the lack of food can inhibit growth. The three times a day of feeding frequency on Asian seabass, *Lates calcarifer* shows better for fish growing [6]. The same result has also been reported that optimal of feeding frequency can increase the growth both length and weight of Tilapia, *Oreochromis niloticus*, [17] and catfish, *Clarias gariepinus* [18].

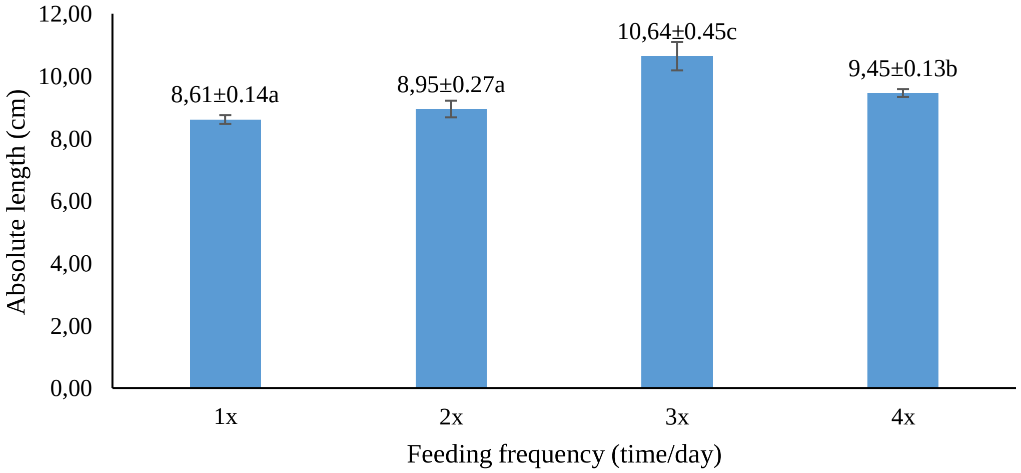


Figure 2. The absolute length of glass eel cultured at different feeding frequency. Bars followed by the same superscript letters are nor significantly different ($P>0.05$).

The absolute weight of glass eel cultured at different feeding frequency shows that the three times a day of feeding frequency is the highest on absolute weight (1.91±0.28g), followed by four times (1.25±0.11g), two times (1.03±0.13g), and one time (0.92±0.10g) a day, respectively (Figure 3). Statistical analysis revealed that three times a day of feeding frequency is significantly different ($P<0.05$) compared to the others on absolute weight. At the present experiment shows that the feeding frequency is also affect the absolute weight where the three times a day is the best for glass eel culture. These are feeding frequencies affects the weight growth have been reported the common carp [10] and Tilapia, *Oreochromis niloticus* [9] at four times a day are the best, Asian seabass, *Lates calcarifer* is at three times a day [6], and Amazon ornamental fish, *Heros severus* is at two times a day [19]. The present experiment shows that three times a day is the best for weight growth of glass eel. The differences of feeding frequency suggesting that different species has different from food whether quality and quantity, besides the types of food such as carnivorous, herbivorous, and omnivorous.

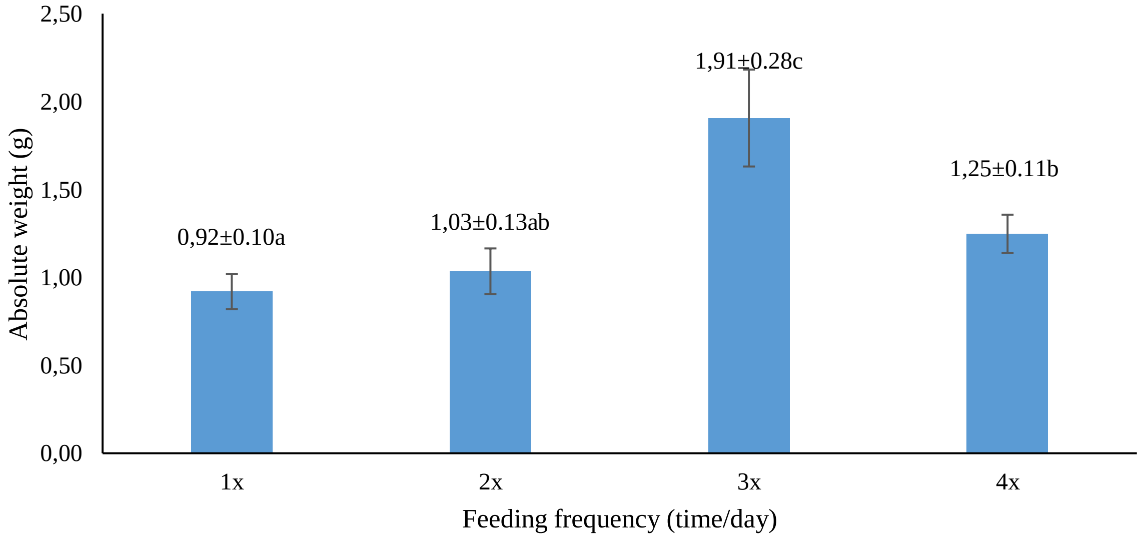


Figure 3. The absolute weight of glass eel cultured et different feeding frequency. Bars followed by the same superscript letters are not significantly different ($P>0.05$)

The high of feeding frequency at the present experiment shows decrease in growth and survival. This may be caused by a lot of food that affect the physiological of fish. Based on the observation, uneaten food is found on the bottom of tank at four time a day of feeding frequency where at the other treatments are relatively clean on the bottom. This indicator that the glass eels are lack of appetite. The feeding frequency affects the amylase activity where the four times a day of feeding frequency has result in decreasing of amylase level even inhibit of amylase activity [10]. The decreasing of amylase level would be followed by loss of appetite. Thus, the present experiment suggesting that at four times a day of feeding rate low of growth and a lot of uneaten food.

3.2. Blood glucose

The result of blood glucose of glass eel cultured at different feeding frequency shows that the blood glucose value has increase with increasing the feeding frequency but decrease at four times a day. The blood glucose value at three times a day is the highest (22.20±3.37 mmol/L), followed by four times (21.05±2.35 mmol/L), two times (12.62 ±2.42 mmol/L), and one time (6.53±1.57 mmol/L) a day, respectively (Figure 4). Statistical analysis revealed that three times a day of feeding frequency is significantly influenced blood glucose value (P<0.05) compered to the other treatments. Blood glucose is not indicator for stress response. At the present experiment shows that increase in feeding frequency led to increase in blood glucose, except for four times a day. The increasing of blood glucose may be related to food intake where a lot of food intake that contains carbohydrate has result in increasing blood glucose due to metabolism process. The same result that blood glucose increase with increasing feeding frequency on red tilapia, *Oreochromis* spp [20].

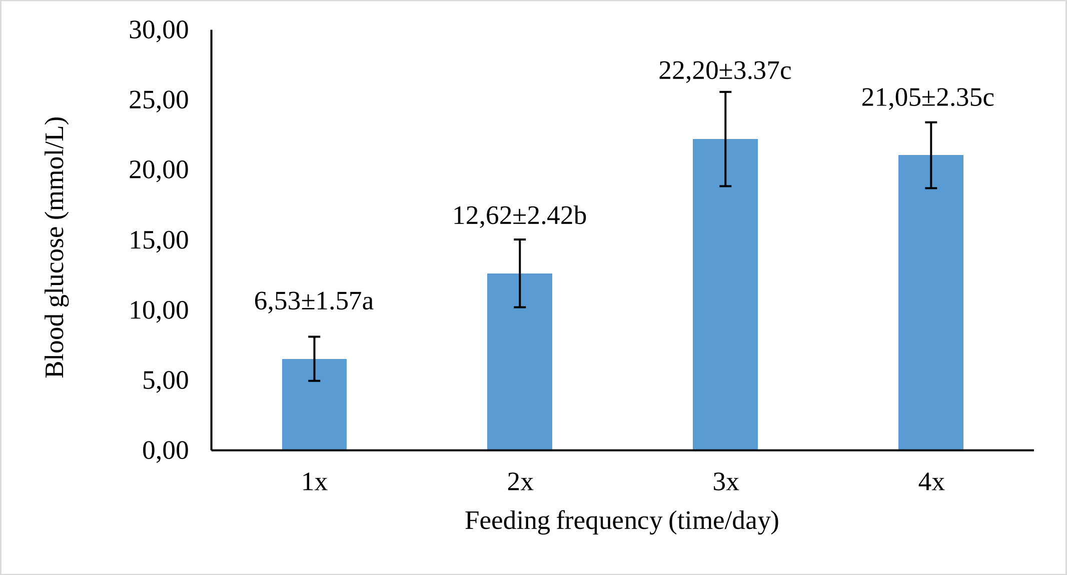


Figure 4. The blood glucose of glass eel cultured at different feeding frequency. Bars followed by the same superscript letters are not significantly different (P<0.05).

3.3 Water quality parameters

Water quality parameters such as temperature, dissolved oxygen, pH, Ammonia (NH3-N), nitrite (NO2-N), nitrate, (NO3-N), Phosphate (PO4-P), and alkalinity is presented at Table 1.

Table 1. The ranged of water quality parameters during the glass eel cultured at different feeding frequency.

Parameters	Unit	Feeding frequency (time/day)			
		1x	2x	3x	4x
Temperature	°C	29.03-30.29	29.07-30.31	29.04-30.25	29.02-30.21
pH		7.24-7.33	7.14-7.27	7.18-7.21	7.20-7.30
DO	mg/L	3.35-4.03	3.27-4.01	3.38-4.05	3.3-4.03
Ammonia	mg/L	0.011-.0.029	0.014-0.033	0.012-0.048	0.011-0.057
Nitrite	mg/L	0.0010-0.0012	0.0010-0.0013	0.0013-0.0017	0.0015-0.0021
Nitrate	mg/L	0.024-0.035	0.022-0.039	0.026-0.041	0.023-0.056
Phosphate	mg/L	0.0121-0.0201	0.0113-0.0214	0.0110-0.0209	0.00124-0.0218
Alkalinity	mg/L	92-102	91-101	91-103	92-101

All data of ranging of the water quality parameters during the culture period of glass eel at different feeding frequency (Table 1) are within the threshold for glass eel culture [21]

4. Conclusion

The feeding frequency affects the survival, growth, and blood glucose on glass eel culture in a recirculating aquaculture system. The increasing of feeding frequency would be led to increase in blood glucose. The highest of feeding frequency has results in loss of appetite and it caused decreased in blood glucose, reduce the growth both of absolute length and absolute weight. The three times a day of feeding frequency is the best in terms of survival, growth, and blood glucose value, therefore, it can be recommended to be applied for glass eel culture in a recirculating aquaculture system.

References

[1] Arai T and Chino N. 2019. Variations in the migratory history of the tropical catadromous eels *Anguilla bicolor bicolor* and *A. bicolor pacifica* in south-east Asian waters. *Journal of Fish Biology*, **94**(5): 752-758

[2] Miller, M. J., Powell, J., Tsukamoto K. 2008. Observation of large metamorphosing Leptochepalus in a coral reef habitat at Sangeang Island, Indonesia. Zoological Studies, **48**: 1-107

[3] Setiadi E., Saputra A., and Taufik I. 2021. The effects of differences in water exchange on the productivity of glass eel stage in *Anguilla bicolor bicolor* culture. E3S Web of Conferences 322, ISFFS 2021, 02006 (2021). https://doi.org/10.1051/e3sconf/202132202006

[4] Nawir F, Utomo NBP, Budiardi T. 2015. Pertumbuhan ikan sidat yang diberi kadar protein dan rasio energi protein pakan berbeda. *Jurnal Akuakultur Indonesia*, **14(**2): 128-134.

[5] Setiadi E., Mulyana, and Fajrian R. A. 2021. The survival rate and growth performance of glass eel (*Anguilla bicolor bicolor*) maintained with different light intensity. *Jurnal Mina Sains***,** **7**(2): 94-103

[6] Hassan H. U., Ali Q. M., Khan W., Masood Z. M., Abdel-Azis M. F. A., Shah M. I. A., Gabol K., Chatta A. M., Kamal M., Zulfikar T., and Hossain M. Y. 2021. Effect of feeding frequency as a rearing system on biological performance, survival, body chemical composition and economic efficiency of Asian seabass *Lates calcarifer* (Bloch, 1790) reared under controlled environmental conditions. *Saudi Journal of Biological Sciences*, 28: 7360-7366

[7] Wang Y., Xie S., Nie Z., Li Q., Sun Y., Shao N., Gao J., Hu J., Xu P., and Xu G. Optimum feeding frequency of juvenile largemouth bass (*Micropterus salmoides*) reared in in-pond raceway recirculating culture system. *Fish Physiol. Biochem.*, **46**:2197–2212

[8] Cadorin D. I., Silva M. F. O., Masagounder K., and Fracalossi D. M. 2022. Interaction of feeding frequency and feeding rate on growth, nutrient utilization, and plasma metabolites of juvenile genetically improved farmed Nile tilapia, *Oreochromis niloticus*. *World Aquaculture Society*,**53**:500-515

[9] Villaroel M., Alavrino J. N. R., and Lopez-Luna J. 2011. Effect of feeding frequency on one day fasting on Tilapia (*Oreochromis niloti*cus) and water quality. *The Israeli of Journal Aquaculture-Bamidgeh*, **63**(609):1-6

[10] Cheng Z., Wang A., Fan Z., Sun J., Cui P., and Qiao X. 2019. Effect of dietary Carbohydrate/Protein ratios and feeding frequency on carbohydrate metabolism of common carp. The 5th Annual International Conference on Material Engineering and Application. IOP Conf. Series: Materials Science and Engineering 484 (2019) 01201. IOP Publishing

[11] Campelo D. A. V., Marques M. H. C., Marim O.P., de Maura L. B. Eiras B. J. C. S., Brabo M. F., and Veras G. C. 2019. Effects of feeding rates and feeding frequencies on growth performance, uniformity of the batch and survival rate of Amazon ornamental fish larvae. *International Journal of Fisheries and Aquaculture*. **11**(2): 23-28

[12] Widyawati A., Sauptra A., and Setiadi E. 2019. Production performance and blood profile of climbing perch, *Anabas testudineus* Bloch cultured in peat pond with different stocking density. *Indonesian Aquaculture Journal*. **14**(2):83-89

[13] Effendie I. 2002. Fisheries Biology. Yayasan Pustaka Nusantara, Yogyakarta. Indonesia

[14] APHA. 1985. Standard method for the examination of water and wastewater. 16th Ed., American Public Health Association, Washinton DC.

[15] Jahanbakhshi A, Imanpoor MR, Taghizadeh V, Shabani A. (2013) Hematological and serum biochemical indices changes induced by replacing fish meal with plant protein (sesame oil cake and corn gluten) in the great sturgeon (*Huso huso*). Comp Clin Pathol., **22**:1087–1092

[16] Silva E., Sterzelecki F. C., Musialak L. A., Sugai J. K., and Cerqueira V. R. 2020. Effect of feeding frequency on growth performance, blood metabolites, proximate composition, and digestive enzymes of lebranche mullet (*Mugil liza*) juveniles. *Aquac. Res.* **51**:1162–1169

[17] Ferdpus Z., Nahar N., Hossen M. S., Sumi K. R., and Ali M. M. 2014. Performance of different feeding frequency on growth indices and survival of monosex tilapia (*Orechromis niloticus*) fry. *Int. J. Fish. Aquat. Stud.,* **1**:80;83

[18] Jamabo N. A., Fubara R. I., and Dienye H. E. 2015. Feeding frequency on growth and feed conversion of (*Clarias gariepinus*) fingerlings. Int. J, Fish. Aquat. Stud., **3**:353-356

[19] Paixao D. J. M. R., Brabo M. F., Soares L. M. O., Campelo D. A. V., and Veras G. C. 2019. Optimal feeding frequency for *Heros severus* (Heckel, 1840) an Amazon ornamental fish. Brazilian Journal of Animal Sciences, 1-7

[20] Robisalmi A., Alipin K., and Gunadi B. 2021. Effect of different feed restrictions on growth, biometric, and hematological response of juvenile red Tilapia (*Oreochromis* spp.). *Jurnal Imialh Perikanan dan Kelautan*. **13**(2):243-254

[21] Setiadi E., Saputra A., and Taufik I. 2021. The effects of differences in water exchange on the productivity of glass eel stage in *Anguilla bicolor bicolor* culture. E3S Web of Conferences 322, 02006 (2021). Conferences 322, 02006 (2021). ISFFS 2021